

## SAFETY DATA SHEET

Creation Date 19-April-2010

Revision Date 29-March-2024

Revision Number 5

### 1. Identification

**Product Name** Hydrazine monohydrate

**Cat No. :** 16651

**CAS-No** 7803-57-8  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Flammable liquids</b>                                | Category 4   |
| <b>Acute oral toxicity</b>                              | Category 3   |
| <b>Acute dermal toxicity</b>                            | Category 3   |
| <b>Acute Inhalation Toxicity</b>                        | Category 2   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 B |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Skin Sensitization</b>                               | Category 1   |
| <b>Carcinogenicity</b>                                  | Category 1B  |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Combustible liquid  
Fatal if inhaled  
Toxic if swallowed or in contact with skin  
Causes severe skin burns and eye damage  
May cause an allergic skin reaction  
May cause respiratory irritation  
May cause cancer

**Precautionary Statements****Prevention**

Obtain special instructions before use  
Do not handle until all safety precautions have been read and understood  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Contaminated work clothing should not be allowed out of the workplace  
Wear protective gloves/protective clothing/eye protection/face protection  
Wear respiratory protection

**Response**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component             | CAS-No    | Weight % |
|-----------------------|-----------|----------|
| Hydrazine monohydrate | 7803-57-8 | >95      |
| Hydrazine             | 302-01-2  | -        |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is

|                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Inhalation</b>                      | If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required.                                                                                                                                                                                                                                                                                |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. May cause allergic skin reaction. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## 5. Fire-fighting measures

|                                         |                                                                                                                                                                                                                |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Use: Dry chemical. Carbon dioxide (CO <sub>2</sub> ). Water spray. Alcohol resistant foam. Water mist may be used to cool closed containers. CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                                                       |
| <b>Flash Point</b>                      | 75 °C / 167 °F                                                                                                                                                                                                 |
| <b>Method -</b>                         | No information available                                                                                                                                                                                       |
| <b>Autoignition Temperature</b>         | No information available                                                                                                                                                                                       |
| <b>Explosion Limits</b>                 |                                                                                                                                                                                                                |
| <b>Upper</b>                            | 99.99%                                                                                                                                                                                                         |
| <b>Lower</b>                            | 3.5%                                                                                                                                                                                                           |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                                                       |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                                                       |

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Combustible material. Containers may explode when heated. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Ammonia. Nitrogen oxides (NO<sub>x</sub>). Hydrogen.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 4             | 2                   | 1                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. Remove all sources of ignition. Take precautionary measures against static discharges. |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.          |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition.                                                                                                                             |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Store under an inert atmosphere. Corrosives area. Incompatible Materials. Strong oxidizing agents. Acids. Metals. copper. Halogens. Peroxides.                                                |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component | Alberta                                              | British Columbia      | Ontario TWAEV         | Quebec                                              | ACGIH TLV             | OSHA PEL                                                                                                           | NIOSH                                                                |
|-----------|------------------------------------------------------|-----------------------|-----------------------|-----------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Hydrazine | TWA: 0.01 ppm<br>TWA: 0.01 mg/m <sup>3</sup><br>Skin | TWA: 0.01 ppm<br>Skin | TWA: 0.01 ppm<br>Skin | TWA: 0.1 ppm<br>TWA: 0.13 mg/m <sup>3</sup><br>Skin | TWA: 0.01 ppm<br>Skin | (Vacated) TWA: 0.1 ppm<br>(Vacated) TWA: 0.1 mg/m <sup>3</sup><br>Skin<br>TWA: 1 ppm<br>TWA: 1.3 mg/m <sup>3</sup> | IDLH: 50 ppm<br>Ceiling: 0.03 ppm<br>Ceiling: 0.04 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection Hand Protection

Goggles  
Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments                                         |
|----------------|-------------------|-----------------|--------------------------------------------------------|
| Nitrile rubber | > 480 minutes     | 0.38 mm         | As tested under EN374-3                                |
| Neoprene       | > 480 minutes     | 0.45 mm         | Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local

conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143 Ammonia and organic ammonia derivatives filter Type K Green conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

#### Hygiene Measures

When using do not eat, drink or smoke. Provide regular cleaning of equipment, work area and clothing.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | Colorless                |
| Odor                                   | Ammonia-like             |
| Odor Threshold                         | No information available |
| pH                                     | 12.0 Alkaline            |
| Melting Point/Range                    | -51.7 °C / -61.1 °F      |
| Boiling Point/Range                    | 118 °C / 244.4 °F        |
| Flash Point                            | 75 °C / 167 °F           |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | 99.99%                   |
| Lower                                  | 3.5%                     |
| Vapor Pressure                         | 10mbar @ 20 deg C        |
| Vapor Density                          | No information available |
| Specific Gravity                       | No information available |
| Solubility                             | Miscible with water      |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | 1.50 mPa s @ 20 deg C    |
| Molecular Formula                      | H4N2.H2O                 |
| Molecular Weight                       | 50.06                    |

## 10. Stability and reactivity

|                                  |                                                                                                                        |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                             |
| Stability                        | Do not allow evaporation to dryness. Air sensitive.                                                                    |
| Conditions to Avoid              | Incompatible products. Excess heat. Exposure to air. Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents, Acids, Metals, copper, Halogens, Peroxides                                                    |
| Hazardous Decomposition Products | Ammonia, Nitrogen oxides (NOx), Hydrogen                                                                               |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                               |

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component             | LD50 Oral                | LD50 Dermal                | LC50 Inhalation                              |
|-----------------------|--------------------------|----------------------------|----------------------------------------------|
| Hydrazine monohydrate | LD50 = 169 mg/kg ( Rat ) | Not listed                 | Not listed                                   |
| Hydrazine             | LD50 = 60 mg/kg ( Rat )  | LD50 = 91 mg/kg ( Rabbit ) | 570 ppm ( Rat ) 4 h<br>0.75 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** Causes severe burns by all exposure routes

**Sensitization** May cause sensitization by skin contact

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component             | CAS-No    | IARC       | NTP                    | ACGIH      | OSHA       | Mexico     |
|-----------------------|-----------|------------|------------------------|------------|------------|------------|
| Hydrazine monohydrate | 7803-57-8 | Not listed | Not listed             | Not listed | Not listed | Not listed |
| Hydrazine             | 302-01-2  | Group 2A   | Reasonably Anticipated | A3         | X          | A3         |

*IARC (International Agency for Research on Cancer)*

*IARC (International Agency for Research on Cancer)*

*Group 1 - Carcinogenic to Humans*

*Group 2A - Probably Carcinogenic to Humans*

*Group 2B - Possibly Carcinogenic to Humans*

*NTP: (National Toxicity Program)*

*NTP: (National Toxicity Program)*

*Known - Known Carcinogen*

*Reasonably Anticipated - Reasonably Anticipated to be a Human Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*A1 - Known Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Animal Carcinogen*

*ACGIH: (American Conference of Governmental Industrial Hygienists)*

*Mexico - Occupational Exposure Limits - Carcinogens*

*Mexico - Occupational Exposure Limits - Carcinogens*

*A1 - Confirmed Human Carcinogen*

*A2 - Suspected Human Carcinogen*

*A3 - Confirmed Animal Carcinogen*

*A4 - Not Classifiable as a Human Carcinogen*

*A5 - Not Suspected as a Human Carcinogen*

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the

hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component | Freshwater Algae                                                                                                                                                                                 | Freshwater Fish                                                                                                                                                                                                                                                                                        | Microtox                                                                     | Water Flea |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------|
| Hydrazine | EC50: = 0.006 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: = 0.071 mg/L, 72h (Pseudokirchneriella subcapitata)<br>EC50: = 0.02 mg/L, 96h static (Pseudokirchneriella subcapitata) | LC50: 0.28 - 1.34 mg/L, 96h static (Poecilia reticulata)<br>LC50: 1.81 - 2.79 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: = 1.17 mg/L, 96h (Lepomis macrochirus)<br>LC50: 0.54 - 1.31 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 0.7 - 1.3 mg/L, 96h flow-through (Lepomis macrochirus) | EC50 = 0.01 mg/L 15 min<br>EC50 = 0.01 mg/L 20 min<br>EC50 = 0.02 mg/L 5 min | Not listed |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

| Component             | log Pow |
|-----------------------|---------|
| Hydrazine monohydrate | 1.1     |
| Hydrazine             | -0.16   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component            | RCRA - U Series Wastes | RCRA - P Series Wastes |
|----------------------|------------------------|------------------------|
| Hydrazine - 302-01-2 | U133                   | -                      |

## 14. Transport information

### DOT

UN-No UN2030  
 Proper Shipping Name HYDRAZINE AQUEOUS SOLUTION  
 Hazard Class 8  
 Subsidiary Hazard Class 6.1  
 Packing Group II

### TDG

UN-No UN2030  
 Proper Shipping Name HYDRAZINE AQUEOUS SOLUTION  
 Hazard Class 8  
 Subsidiary Hazard Class 6.1  
 Packing Group II

### IATA

UN-No UN2030  
 Proper Shipping Name HYDRAZINE, AQUEOUS SOLUTION  
 Hazard Class 8

|                         |                             |
|-------------------------|-----------------------------|
| Subsidiary Hazard Class | 6.1                         |
| Packing Group           | II                          |
| <b>IMDG/IMO</b>         |                             |
| UN-No                   | UN2030                      |
| Proper Shipping Name    | HYDRAZINE, AQUEOUS SOLUTION |
| Hazard Class            | 8                           |
| Subsidiary Hazard Class | 6.1                         |
| Packing Group           | II                          |

## 15. Regulatory information

**All of the components in the product are on the following Inventory lists:** China X = listed Australia U.S.A. (TSCA) Canada (DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS)

### International Inventories

| Component             | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Hydrazine monohydrate | 7803-57-8 | -   | -    | -    | -                                             | -         | -      | -   |
| Hydrazine             | 302-01-2  | X   | -    | X    | ACTIVE                                        | 206-114-9 | -      | -   |

| Component             | CAS-No    | IECSC | KECL       | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------------|-----------|-------|------------|------|------|------|------|-------|-------|
| Hydrazine monohydrate | 7803-57-8 | X     | KE-05-0709 | X    | -    | X    | X    | X     | X     |
| Hydrazine             | 302-01-2  | X     | KE-19981   | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA)         |
|-----------|------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------|
| Hydrazine | Part 1, Group B Substance                            | Schedule I                                                                 | Subject to Monitoring and Surveillance Activities |

#### Legend

NPRI - National Pollutant Release Inventory

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component             | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                   | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Hydrazine monohydrate | -                                                                   | -                                                                                               | SVHC Candidate list - 616-584-0 - Carcinogenic, Article 57a                                           |
| Hydrazine             | -                                                                   | Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 75. | SVHC Candidate list - 206-114-9 - Carcinogenic, Article 57a                                           |

|  |  |                                    |  |
|--|--|------------------------------------|--|
|  |  | (see link for restriction details) |  |
|--|--|------------------------------------|--|

After the sunset date the use of this substance requires either an authorization or can only be used for exempted uses, e.g. use in scientific research and development which includes routine analytics or use as intermediate.

**REACH links**

<https://echa.europa.eu/authorisation-list>

<https://echa.europa.eu/substances-restricted-under-reach>

<https://echa.europa.eu/candidate-list-table>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component             | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Hydrazine monohydrate | 7803-57-8 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Hydrazine             | 302-01-2  | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component             | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Hydrazine monohydrate | 7803-57-8 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Hydrazine             | 302-01-2  | 0.5 tonne                                                                                 | 2 tonne                                                                                  | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date**

19-April-2010

**Revision Date**

29-March-2024

**Print Date**

29-March-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**